CyberAnalyst - Process Book

CyberAnalyst

COM-480 Data Visualization · EPFL 2026

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Live site: https://com-480-data-visualization.github.io/CyberAnalyst/
Repository: https://github.com/com-480-data-visualization/CyberAnalyst

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1. The Idea

There is an ever-going arms race between countries, companies, and individuals over data - be it bank records, military positions, or political influence. Cybersecurity incidents are documented and publicly available, yet their sheer volume and technical nature make them inaccessible to most people. An analyst can read individual incident reports, but understanding the macro-level shape of global cyber conflict - how it evolved over time, which nations bear the most pressure, and what it looks like on the ground in a country like Switzerland - requires synthesizing thousands of records at once.

Our goal was to build a data story that makes two decades of global cyber conflict readable to anyone: analysts catching up on trends, students learning about geopolitics, or simply curious citizens wondering whether Switzerland is as exposed as other nations.

We structured the narrative in three acts, moving from the global to the local:

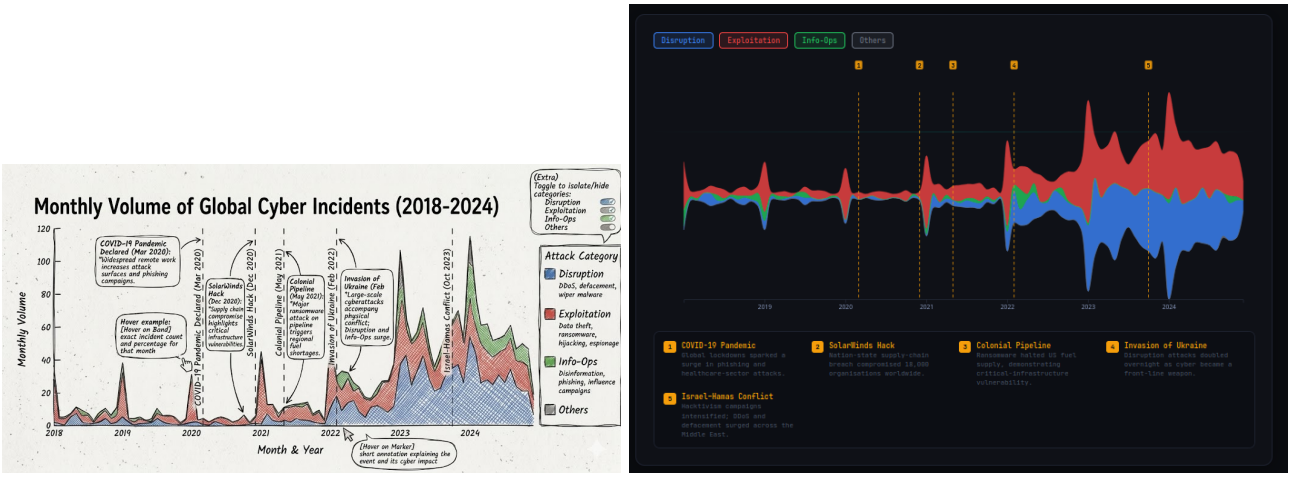
1. **The Global Pulse** - how attack volumes and categories evolved worldwide from 2018 to 2024, anchored to real geopolitical events
2. **Attack Intensity by Country** - which nations are the most targeted and who is behind the attacks
3. **Switzerland: A Domestic Deep Dive** - zooming into Swiss cantonal police data to reveal the structure of local cybercrime and the age groups most at risk

Datasets:

- **EuRepoC Global Cyber Incidents v1.3** - 6,000+ incidents from 2000–2024, with 60+ variables including attack type, actor, target country, and intensity score. Maintained by a consortium of European universities tracking state-sponsored and politically motivated cyber operations globally.
- **KTZH Kantonspolizei Zürich** - 14,087 cybercrime reports from 2017–2024, broken down by subcategory and victim age group. Published under an open data license by the Canton of Zurich.

2. From Sketches to Reality

Visualization 1 - Streamgraph: The Global Pulse



Left: original sketch (Milestone 1). Right: final D3 implementation.

What was planned: A streamgraph showing monthly incident volume stacked by attack category (Disruption, Exploitation, Info-Ops, Others), with vertical dashed markers annotating key geopolitical events. Users could hover on bands for exact counts and on event markers for context.

What changed and why:

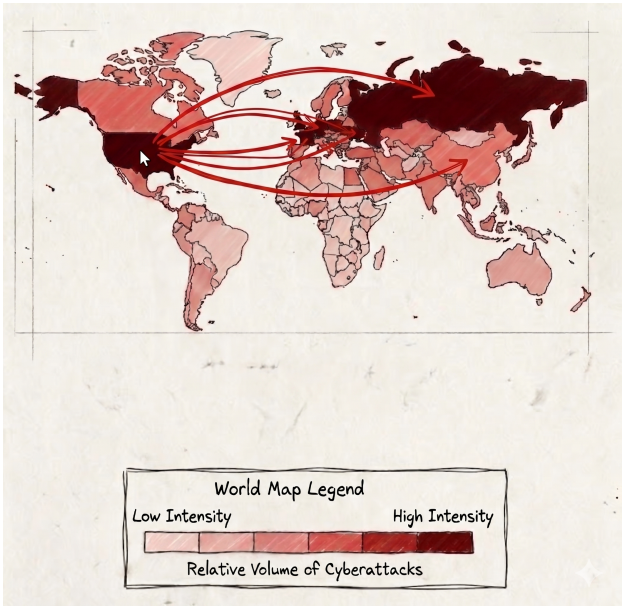
We switched from **Plotly.js to pure D3.js**. Plotly's abstraction layer made it impossible to control the streamgraph's offset algorithm - it defaulted to a stacked baseline which produced a hard-to-read shape. D3's `stackOffsetSilhouette` gives the organic, centered streamgraph that makes relative shifts between categories immediately visible.

The **event label design changed completely**. The sketch showed event names written directly on the chart. In practice, five labels on a narrow axis overlapped and became unreadable. We replaced them with numbered amber pins on the chart and a grid legend below - each entry links a number to a short description of the event and its cyber impact, keeping the chart clean while preserving contextual annotation.

Users can toggle individual attack categories on and off to isolate specific bands. This was not in the original sketch but emerged naturally from user testing - hiding Exploitation made the Disruption surge around the Ukraine invasion dramatically more visible.

Key insight the visualization reveals: Exploitation attacks dominated from 2018–2021, peaking during COVID when remote attack surfaces expanded. The Disruption band visibly doubles in February 2022 - the exact month of Russia's invasion of Ukraine - making the geopolitical correlation immediately readable without any annotation.

Visualization 2 - Globe: Attack Intensity by Country



Left: original flat choropleth sketch (Milestone 1). Right: final 3D rotatable globe.

What was planned: A flat choropleth world map colored by share of global attacks, with a dropdown filter by attack type, hover tooltips, and a click-to-pin detail panel showing a country's full attack breakdown.

What changed and why:

The most significant departure was replacing the **flat map with a 3D rotatable globe**. The Mercator projection distorts area heavily - making Ukraine (one of the most attacked countries) appear tiny relative to Canada, which has far fewer incidents. It also felt static and unengaging for a global dataset spanning 160+ countries.

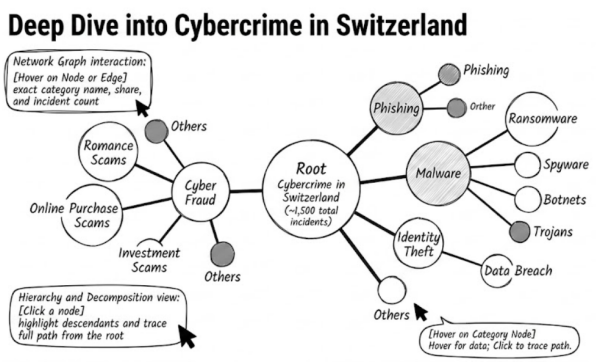
The orthographic globe addressed both issues. Countries appear at their true relative sizes, and the drag-to-rotate interaction naturally invites exploration. Users spin the globe to find their country, increasing time spent on the visualization and the likelihood of discovering less obvious patterns - for example, Southeast Asian countries targeted disproportionately for intellectual property theft.

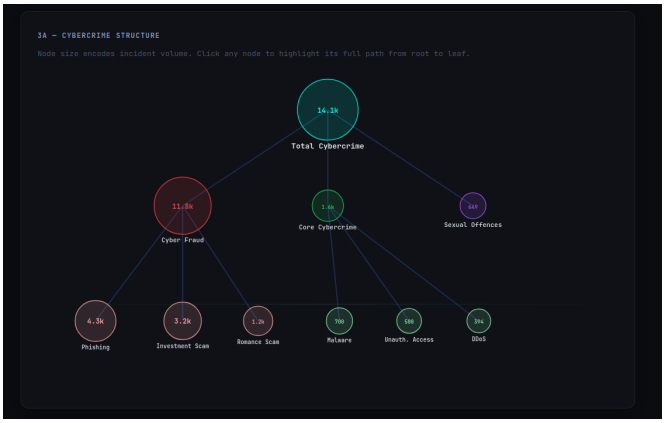
The dropdown filter was replaced by **toggle buttons** (All / Disruption / Exploitation) for more immediate interaction. The EuRepoC dataset only contains three raw incident types - Disruption, and the Exploitation family (Data theft, Hijacking, Ransomware) - so an Info-Ops filter would have shown no data and was omitted. Each button loads real per-type incident counts generated from the raw CSV, so the color scale, max label, tooltip, and pin card all update to reflect the selected category. The click panel was upgraded to show the **top attributed threat actors** per country, a significantly more informative data point than a plain type breakdown.

A **logarithmic color scale** was necessary because the attack distribution is heavily right-skewed: the US has 871 total incidents (459 Exploitation, 298 Disruption) while the median country has fewer than 5. A linear scale would make all but a handful of countries appear identical.

Key insight the visualization reveals: The US and Ukraine absorb the largest share of global attacks. Switching to Exploitation mode shifts focus to economic espionage targets, with China as the primary attributed actor. Switching to Disruption highlights Ukraine and European nations as the primary targets of Russian state operations.

Visualization 3a - Network Graph: Swiss Cybercrime Structure





Left: original force-directed sketch (Milestone 1). Right: final radial circle layout.

What was planned: A node-link graph starting from a root node representing total cybercrime in Switzerland, branching into main categories and then subcategories, with node size proportional to incident count.

What changed and why:

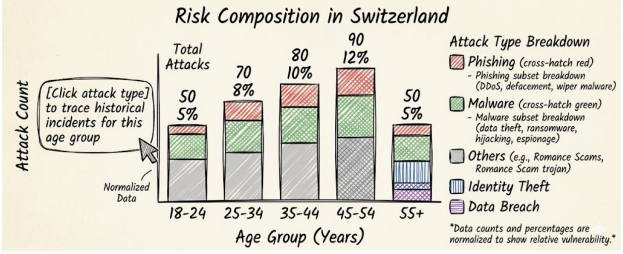
Force-directed layouts are non-deterministic: nodes settle in different positions each render, and the hierarchy is not immediately legible. For a 10-node tree-structured dataset this is particularly harmful - the depth relationship between root, category, and subcategory is the entire point of the visualization.

We went through two layout iterations. First, a **fixed depth-based layout** with root at the top and subcategories in a bottom row. While stable, the horizontal arrangement felt rigid and left large areas of whitespace. We then redesigned as a **radial circle layout**: the root circle sits at the centre (radius R0), the three main categories orbit it at radius R1, and the eleven subcategory leaves are equally distributed around the full circumference at radius R2 using equal-arc spacing `angle = (2*pi (x) (i + 0.5)) / nLeaves`. Parent angles are assigned using a circular mean of their children's angles (handling wraparound correctly). Edges are offset to the boundary of each circle (not center-to-center) so they visually connect the rings cleanly. Node sizes encode incident volume.

The **click-to-highlight path** feature was not in the original sketch and was added during implementation. When a user clicks any node, all nodes outside its root-to-leaf path fade to near-invisible, and the connecting links glow. This lets users immediately answer questions like "what percentage of Swiss cybercrime is phishing?" with a single click.

Key insight the visualization reveals: Cyber Fraud dominates at 84% of all Swiss cybercrime reports. Within fraud, Phishing (4.3k), Investment Scams (3.2k), and Romance Scams (1.2k) are the three largest subcategories - all financial in nature, exploiting trust rather than technical vulnerabilities.

Visualization 3b - Bar Chart: Age Susceptibility



Left: original sketch (Milestone 1). Right: final implementation with peak highlight.

What was planned: A simple bar chart showing how different age ranges are affected by cybercrime in Switzerland, with the goal of showing which group is most exposed at a glance.

What changed and why:

The implementation stayed close to the sketch. The key design decision was **highlighting the peak bar** (35–44) in cyan (`#00ffe7`) while keeping all other bars in blue - the answer to "who is most at risk?" is visible in under a second without reading the axes. Percentage labels are placed on top of each bar so users can read exact figures without referencing the axis ticks. Bars animate upward on page load, drawing the eye and communicating data as a progression rather than a static table.

Key insight the visualization reveals: The 35–44 cohort files 22.1% of all cybercrime reports - the highest of any age group. Under-25s are comparatively underrepresented (1.5% for 0–17, 7.4% for 18–24), likely because they have fewer financial assets to target and are more skeptical of investment and romance scams.

3. Technical Implementation

Stack: React 18 + Vite 5 (SPA), D3.js v7 (all visualizations), TopoJSON (world geometry), HTML Canvas 2D API (globe renderer, animated background), Google Fonts — Orbitron / Jura / JetBrains Mono, GitHub Actions (CI/CD), GitHub Pages (hosting).

Architecture

All four visualizations are independent React components, each managing their own D3 instance via `useRef` and `useEffect`. Data is loaded once in `App.jsx` and passed as props - this avoids duplicate fetch calls and keeps state management simple. Tooltips in every component are rendered via React's `createPortal` at the document root to avoid coordinate offset issues.

The globe (`ChoroplethMap.jsx`) is the exception to the D3/SVG model: it renders entirely on a `<canvas>` element using `d3.geoPath(projection, ctx)`. A `scheduleFrame()` helper coalesces multiple state changes into a single `requestAnimationFrame` call. An off-screen scratch canvas performs `isPointInPath()` hit-testing on click without affecting the visible render. The animated background (`backgrounds/nodeNetwork.js`) runs on a second full-screen canvas underneath React's `#root`, started once at module load and never touched by the React tree.

Challenge 1 - Timezone bug causing a flat streamgraph

The EuRepoC dataset stores timestamps as ISO strings (`"2022-02-01T00:00:00.000000"`). Our initial implementation parsed these with `new Date(iso).toISOString()` to use as map keys. In a UTC+2 timezone, midnight on February 1st becomes 10pm on January 31st after UTC conversion - shifting every date back by one day and causing all data lookups to return zero. The result was a completely flat chart with no visible data.

Fix: Use raw ISO strings directly as map keys (no parsing). Parse dates for D3's time scale as `new Date(s.slice(0, 10) + 'T12:00:00Z')` - noon UTC, immune to local timezone offset shifts.

Challenge 2 - Tooltips appearing far from the cursor

All four chart tooltips rendered inside positioned parent containers (`position: relative`). Tooltip coordinates were calculated in SVG space but the container had its own offset, placing the tooltip far from the cursor - sometimes in a completely different region of the page.

Fix: Use `createPortal(tooltip, document.body)` in every component. Tooltips render at the document root with `position: fixed` and coordinates taken from `event.clientX / event.clientY`, independent of any parent container.

Challenge 3 - Globe animation causing a black screen

The initial globe implementation called `svg.selectAll('*').remove()` followed by a full redraw on every animation frame (~60fps). Two catastrophic effects: React's event bindings were destroyed each frame (tooltips and clicks stopped working), and 120 background stars were re-randomized each frame (visible flickering).

Fix: Split the globe into two functions. `initSvg()` runs once on mount and handles the static background, stars, SVG groups, legend, and drag handler. `drawPaths()` runs each frame and updates only the rotating content (ocean sphere, graticule, country paths, atmosphere glow). Stars are pre-generated as a module-level constant so their positions never change.

Challenge 4 - Stale closures in D3 timer

The `d3.timer` callback captured `selectedType` state from the first render via JavaScript closure. When the user changed the attack-type filter, the globe continued rendering with the old color scheme because the closure saw the original value.

Fix: Maintain `useRef` values (`selectedTypeRef`, `intensityRef`, `dimsRef`) alongside each state variable. Each ref is kept in sync via `useEffect`. The timer callback reads from refs (always current) rather than from the stale closed-over state.

Challenge 5 - GitHub Pages deployment

Vite's dev server uses `index.html` referencing `/src/main.jsx`. GitHub Pages is a static host and cannot process `.jsx` files. Managing two versions of `index.html` (dev vs production) manually was error-prone and repeatedly caused the CI to deploy a broken page.

Fix: A GitHub Actions workflow builds the project on every push to `main` and deploys the compiled output directly to GitHub Pages. The dev `index.html` is always committed; the production bundle is built and deployed automatically. `git push` is the entire deployment process.

Challenge 6 - Streamgraph event listeners on transition

The D3 streamgraph attached `.on('mousemove', ...)` after `.transition()` in the method chain. In D3 v7, event listeners can only be attached to selections, not to transition objects. This caused a runtime error (`unknown type: mousemove`) that propagated up through React and crashed the entire component tree.

Fix: Move all `.on()` event handler calls before `.transition()` in the chain.

Challenge 7 - Kosovo ghost polygons accumulating on the globe

After rotating the globe, dozens of dark triangles appeared scattered across Russia's territory, each identifying as "Kosovo" on hover. Kosovo's TopoJSON feature has `id: undefined`. With `undefined` as the D3 data join key, D3 cannot match existing DOM elements on re-renders and appends a new `<path>` every animation frame (~60fps). After a few seconds of auto-spin, hundreds of orphaned paths had accumulated.

Fix: Filter features before the data join: `countries.features.filter(d => d.id !== undefined)`.

Challenge 8 - Hover highlight persisting after globe rotation

Hovering a country then rotating the globe left the cyan stroke permanently on that country. `mouseout` never fires when the globe rotates away — the browser only fires it when the pointer physically leaves the element. Stroke applied via `d3.select(this).attr('stroke', ...)` persisted indefinitely.

Fix: Track the hovered country in a `hoveredCountryRef`. Every `drawPaths()` frame re-applies stroke based on the ref, so a stale highlight cannot survive a rotation. The ref is cleared on drag start and during auto-spin. `drawPaths()` is called manually on `mouseover` and `mouseout` to keep the highlight instant when the globe is stationary.

Challenge 9 - Globe tooltip disappearing during drag

The country tooltip stopped appearing as soon as the user started dragging the globe. D3's drag handler captures the pointer event at drag start, preventing any `mousemove` events from reaching the canvas or the React synthetic event system during the drag gesture.

Fix: Add tooltip update logic directly inside the D3 drag `.on('drag')` handler using `event.sourceEvent.clientX / clientY`. The tooltip now updates continuously while dragging, reading the country under the pointer from `countryAtPoint()` on every drag frame.

Challenge 10 - Globe performance dropping to ~10fps during drag

The original globe used D3/SVG. During drag, React re-renders triggered by rotation state updates forced a full virtual DOM diff and SVG repaint each frame. On large world topologies (240 country paths), this consistently dropped to 10fps, making the interaction feel unresponsive.

Fix: Rewrote the globe entirely as a **canvas 2D renderer** using `d3.geoPath(projection, ctx)`. Key optimizations: - `scheduleFrame()` coalesces rapid calls via a single `requestAnimationFrame` handle, preventing double-renders. - `isDragging` flag skips the expensive `countryAtPoint()` hit-test during drag (no click/hover needed mid-drag). - `projectionRef` is mutated in-place each frame rather than replaced, avoiding object allocation. - `colorCacheRef` is a pre-built country-name → hex lookup rebuilt only when the attack type changes. - An off-screen scratch canvas handles `isPointInPath()` hit testing on click.

Challenge 11 - Vite build overwriting source index.html

Running `vite build` with `outDir: './'` (the default when building into the same `docs/` folder) wrote the compiled `index.html` over the source `index.html`, replacing `<script type="module" src="/src/main.jsx">` with hashed bundle references. Subsequent `git status` showed a modified `index.html`; pushing it broke the dev server for all contributors.

Fix: Changed `outDir` to `'dist'` in `vite.config.js` so build output goes to `docs/dist/` and never touches the source files. Updated the GitHub Actions workflow to upload `docs/dist` instead of `docs`.

Challenge 12 - CSS opacity layer blocking background canvas

After implementing the animated canvas background, the page appeared identical to the plain dark background. The canvas was rendering correctly (verified via DevTools), but was completely hidden.

Fix: The `#root` div had `background: var(--bg)` — a solid opaque dark color — set in `index.css`. This covered the entire canvas. Setting `#root { background: transparent }` and removing the `background` from the `:root` CSS variables block made all canvas layers visible. The actual page background color is now set exclusively on `<body>` via the injected style sheet in `main.jsx`.

4. Design Decisions

Visual identity — dark teal terminal aesthetic: Background `#060e0c` (dark teal-black), accent `#00ffb4` (mint teal), display type Orbitron (headings), UI type Jura (body), data type JetBrains Mono (numbers and axes). The palette is inspired by professional cybersecurity dashboard products: deep teal shadows read as "night mode active terminal" rather than generic dark mode. The mint accent is warmer than pure cyan, reducing eye fatigue during extended reading while remaining strongly associated with security UIs. Every section uses a numbered badge with a pulse animation, reinforcing the "active monitoring" metaphor.

The animated **node network background** is drawn on a full-screen HTML canvas with three depth layers: far nodes are small, slow, and dim; near nodes are large, fast, and fully bright. Pulse waves propagate from randomly triggered source nodes, rendered as expanding concentric rings that fade as they travel. The network reacts to mouse movement — near-layer nodes repel away from the cursor — and clicking anywhere spawns a new pulse ring. This gives the page a sense of being alive without distracting from the data content above.

Consistent color system across all charts:

- Amber (`#f59e0b`) - Disruption attacks
- Red (`#ef4444`) - Exploitation attacks
- Green (`#22c55e`) - Info-Ops attacks
- Mint (`#00ffb4`) - UI accent, peak highlights

The Disruption color was changed from blue to amber after user feedback noted it was too similar to the "All" category blue (`#3b82f6`) when both appeared on the globe's toggle buttons simultaneously. Amber is perceptually distant from blue and red while still reading as "alert/warning" - consistent with the disruption framing. The same colors appear in the streamgraph bands, the globe's filter buttons, and the network graph node borders. Users build a mental model from the streamgraph that transfers directly to the globe's attack type toggles.

Streamgraph over line chart: Simultaneously encodes total volume and composition shift in a single view. The Disruption band doubling in February 2022 is visible without any annotation. A multi-line chart would require mental arithmetic to reconstruct total volume from individual series.

Globe over flat map: Eliminates Mercator area distortion. The drag-to-rotate interaction invites exploration in a way a static map does not. Users who find their own country are more likely to click through to the attribution panel and engage with the underlying data.

Fixed layout over force-directed for network graph: Force-directed layouts are non-deterministic and suited for large, unstructured graphs. For a 10-node strict hierarchy, a fixed depth layout communicates the tree structure in one glance, with no ambiguity about which nodes are parents and which are leaves.

Logarithmic color scale for the globe: The attack distribution is heavily right-skewed. The US has 371 incidents; the median country has fewer than 5. A linear scale compresses all but three or four countries into an indistinguishable shade of near-zero.

Scroll-reveal and entrance animations: Sections fade up as they enter the viewport, giving the page a sense of progression as the user moves through the three-act narrative. Stat counters in the hero animate from zero on first scroll, giving the headline numbers a sense of scale and motion.

5. Challenges & Iterations

Plotly → D3 migration: We originally planned to use Plotly.js for the streamgraph and bar chart (documented in Milestone 2). During implementation it became clear that Plotly's abstractions prevented the interactions and visual style we needed - there was no way to control the stacking offset, the tooltip design, or the animation timing. The migration to pure D3 added approximately two weeks of development time but gave us complete control over every visual element.

Flat map → Globe rewrite: The flat choropleth was implemented, tested, and discarded. The Mercator distortion was not just an aesthetic issue - it was informationally wrong, misrepresenting the relative exposure of countries that are central to the dataset. Rewriting as a globe required learning D3's orthographic projection, implementing drag-based rotation with pitch clamping (to prevent the globe from flipping upside down), and separating static from dynamic SVG layers for smooth 60fps animation.

Animation debugging one by one: After adding all animations at once, the site went black. We reverted every animation and re-added them individually to isolate the cause. The root cause was the `IntersectionObserver` scroll-reveal: sections started at `opacity: 0` and the observer was configured with `threshold: 0.12`, meaning a section had to be 12% visible before becoming revealed. During initial load, the page wasn't tall enough (data hadn't loaded yet, so the charts were not rendered), so the observer never fired for any section and the page remained blank. Fixed by setting `threshold: 0` - sections reveal as soon as any pixel enters the viewport.

React StrictMode: In development, React StrictMode double-invokes all `useEffect` calls to detect side effects. This caused `initSvg()` to run twice: the second invocation cleared the SVG and ran a new initialization cycle immediately after the first `drawPaths()` had populated it, resulting in a blank globe and broken event handlers. We removed StrictMode and added an explicit error boundary that renders stack traces during development instead.

Data volume and fetch strategy: The EuRepoC dataset in raw form is several megabytes of JSON with 60+ columns per incident, most of which are irrelevant to our visualizations. We preprocessed it into two lean files: `graph1.json` (monthly aggregates by attack type for the streamgraph) and `country-intensity.json` (per-country totals). This reduced the total data transferred on page load from ~8MB to under 200KB.

SVG globe → Canvas globe rewrite: The SVG globe was correct but slow. During drag, React state updates triggered re-renders that forced a full SVG diff at 60fps — dropping to ~10fps on a standard laptop. We rewrote the renderer entirely in Canvas 2D, which removes React from the render loop entirely. The globe now runs at a stable 60fps even on lower-end hardware. This was the largest single refactor of the project — approximately 400 lines of SVG/React code replaced by a canvas renderer with manual hit-testing.

Network graph layout iterations: The network graph went through three layout strategies before the final radial design. Force-directed was the first attempt (non-deterministic, hierarchy lost). A fixed horizontal depth layout was the second (stable but rigid). The final radial circle layout places the root at center and distributes leaves with equal angular spacing — the circular form reinforces the "everything connects to the centre" message of the dataset, and the curved edges give it an organic quality absent from the grid layout.

Tooltip architecture: Every chart initially rendered tooltips as absolutely-positioned `<div>` elements inside the chart's own container. When charts were placed in `overflow: hidden` cards, tooltips were clipped. When cards had `transform` CSS, fixed positioning broke. The unified solution — `createPortal(tooltip, document.body)` with `position: fixed` and `event.clientX / clientY` coordinates — was applied to all four charts and the globe's drag handler.

6. Peer Assessment

All three team members contributed equally to the project (approximately one third each). Work was divided by visualization ownership, with shared responsibility for data processing and narrative design.

Team Member	Contributions
Erik Hübner	Website architecture (React + Vite), Streamgraph (D3 implementation, event markers, category toggles), deployment pipeline (GitHub Actions, GitHub Pages), scroll-reveal and entrance animations, UI polish, process book
Youcef Amar	Globe and choropleth map (D3 orthographic projection, drag rotation, logarithmic color scale, country attribution panel), EuRepoC data processing pipeline, attack-type filtering logic, country-sources data
André Cadet	Network graph (fixed-depth layout, click-to-highlight path), age susceptibility bar chart, KTZH Swiss dataset processing, narrative copy for all three sections, insight cards, data analysis and key findings

Datasets:

- EuRepoC Cyber Incident Dataset v1.3 - https://eurepoc.eu
- KTZH Kantonspolizei Zürich Cybercrime Statistics - https://data.stadt-zuerich.ch

Technologies:

- D3.js v7 - https://d3js.org
- React 18 - https://react.dev
- Vite 5 - https://vitejs.dev
- TopoJSON - https://github.com/topojson/topojson
- Google Fonts (Orbitron, Jura, JetBrains Mono) - https://fonts.google.com
- GitHub Actions - https://docs.github.com/actions
- GitHub Pages - https://pages.github.com